

Reality Representation Standard - (RRS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space: Simulation Reality Representation

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Canonical Language: English (UCL™)

Governed Space

Simulation Reality Representation

Canonical Definition

Reality Representation Standard (RRS) defines the governance conditions under which the aspects of reality relevant to a governed Simulation Object, Purpose, Scope, Exclusions, Applicability Context, and required Simulation Depth & Precision are selected, abstracted, represented, resolved, simplified, bounded, and documented within a simulation.

RRS establishes the governed relationship between:

Operational Reality

and:

Simulated Representation

so that a simulation cannot silently treat an incomplete, simplified, abstracted, or selectively represented model of reality as though it were equivalent to reality itself.

Core Question

Which aspects of reality must the simulation represent, at what level of abstraction, fidelity, and resolution, and which aspects may legitimately remain simplified, approximated, or unrepresented?

Purpose

The purpose of RRS is to establish how relevant reality is transformed into a governed simulation representation.

The preceding SIMULOS® standards establish:

What is being simulated?

and:

Why, within what scope, and at what depth must it be simulated?

RRS addresses the next foundational question:

Which parts of reality must now be represented inside the simulation, and how must that representation remain connected to the reality it claims to stand for?

A simulation cannot represent all aspects of reality.

Selection, abstraction, simplification, approximation, and resolution are unavoidable.

RRS makes those transformations governable.

Operational Role

RRS is the third Parent Standard of SIMULOS® — Simulation Governance Architecture.

It receives: a governed Simulation Object, a defined Simulation Purpose, a governed Simulation Scope, explicit exclusions, an Applicability Context, and required Simulation Depth & Precision.

It transforms those upstream conditions into a governed Reality Representation Basis suitable for subsequent Simulation Assumptions, Model & Environment, Inputs & Data, Scenario Design, and Simulation Execution.

RRS prevents: selective representation without disclosure, material reality dimensions disappearing during modeling, abstraction being mistaken for equivalence, simplified representation being presented as complete reality, false fidelity, insufficient resolution, over-resolution without methodological value, hidden representation limitations, and simulation outputs being interpreted as though the simulated representation and operational reality were identical.

Governance Flow

Governed Simulation Inquiry → Relevant Reality Identification → Reality Selection → Representation Abstraction → Representation Fidelity →

Representation Resolution → Simplification & Approximation → Representation
Limitations → Reality Representation Basis

The resulting Reality Representation Basis becomes an input to:

Simulation Assumption Governance

and subsequently to governed construction of the:

Simulation Model & Environment Reality Representation

Reality Representation is the governed transformation through which selected characteristics, structures, behaviors, interactions, conditions, relationships, and dynamics of reality become representable within simulation.

Representation may be: physical, mathematical, digital, statistical, behavioral, logical, probabilistic, agent-based, process-based, hybrid, conceptual, or another methodologically appropriate form.

RRS does not prescribe one representation technology.

It governs whether the representation is suitable for the simulation inquiry.

Reality ≠ Representation

SIMULOS® establishes a foundational distinction:

Reality ≠ Representation

A representation may reproduce selected properties of reality.

It does not become reality itself.

Therefore:

Simulation Representation must remain distinguishable from Operational Reality throughout the simulation lifecycle. Representation ≠ Simulation Object

The Simulation Object identifies:

what is being simulated.

Reality Representation determines:

which characteristics of that object and its relevant reality are represented inside the simulation.

A complete Simulation Object may legitimately have a partial representation where the Simulation Purpose requires only selected characteristics.

That partiality must remain explicit.

Reality Selection

Not every characteristic of reality can or should be represented.

Reality Selection determines which reality dimensions are materially relevant to the governed Simulation Purpose and Scope.

Selected reality may include: physical characteristics, system structures, behaviors, environmental conditions, operational states, interactions, dependencies, human behavior, autonomous behavior, temporal dynamics, failure conditions, resource conditions, or other simulation-relevant realities.

Material Reality Element

A Material Reality Element is a characteristic, relationship, behavior, condition, or dynamic whose omission or distortion could materially affect: simulation behavior, simulation outputs, scenario response, reality correlation, interpretation, or downstream reliance.

Material Reality Elements require explicit representation treatment.

Reality Selection Must Derive From Scope

RRS establishes:

Reality selection must follow the governed Simulation Purpose, Scope, Exclusions, Applicability Context, and Depth requirements rather than the convenience of the available model.

A representation tool must not silently define which parts of reality are considered relevant.

Selective Representation

Selective representation is legitimate.

A simulation may intentionally represent only a subset of reality.

However:

Selective Representation ≠ Complete Reality Representation

The selected subset must remain visible.

Representation Relevance

A selected reality characteristic should have a clear relationship to the Simulation Purpose.

Representation relevance helps prevent simulation complexity from increasing simply because more real-world detail can technically be modeled.

Reality Abstraction

Reality Abstraction is the governed reduction of real-world complexity into a form suitable for simulation.

Abstraction may: aggregate components, simplify behavior, reduce dimensionality, remove irrelevant detail, combine states, approximate relationships, or represent complex phenomena through higher-level constructs.

Abstraction is not inherently a weakness.

It becomes problematic when its effect on simulation meaning is hidden.

Abstraction Level

Different elements of the same simulation may legitimately use different abstraction levels.

For example:

Primary control behavior → detailed representation

while:

non-material infrastructure → higher abstraction

may be appropriate.

RRS therefore permits differentiated abstraction.

Abstraction Must Preserve Material Behavior

SIMULOS® establishes:

An abstraction must preserve the characteristics materially necessary for the Simulation Purpose.

If abstraction removes the behavior the simulation is intended to examine, the representation becomes methodologically insufficient.

No Abstraction Concealment Rule

A simulation must not present abstracted behavior as though every underlying real-world mechanism had been represented.

Representation Fidelity

Representation Fidelity describes the degree to which a simulated representation reproduces the characteristics of reality material to the simulation inquiry.

Fidelity may concern: physical behavior, system behavior, interaction behavior, temporal dynamics, environmental behavior, organizational behavior, human behavior, autonomous behavior, or another simulation-relevant dimension.

Fidelity Is Purpose-Relative

There is no universally correct fidelity level.

A representation may be sufficiently faithful for:

concept exploration

while insufficient for:

high-consequence operational assessment.

Fidelity must therefore be evaluated relative to:

Purpose + Scope + Applicability + Depth

Fidelity ≠ Complexity

A more complex representation is not automatically more faithful.

Additional detail may: introduce unsupported assumptions, increase uncertainty, create unnecessary dependencies, or obscure important behavior.

RRS establishes:

Representation complexity does not itself demonstrate Representation Fidelity.
Fidelity ≠ Precision

A representation can be highly precise while poorly representing reality.

Likewise, a lower-precision representation may adequately reproduce the behavior material to the inquiry.

Therefore:

Precision must not be used as a proxy for fidelity.

Fidelity Differentiation

Different representation elements may require different fidelity levels.

For example:

Spacecraft Guidance Behavior → High Fidelity

Non-Material Visual Appearance → Low Fidelity

where visual appearance does not affect the simulation purpose.

RRS allows fidelity resources to follow materiality.

Representation Resolution

Representation Resolution defines the level at which the simulated representation distinguishes relevant characteristics, states, entities,

events, spaces, times, and relationships.

Resolution may be: spatial, temporal, behavioral, structural, population-based, state-based, event-based, or interaction-based.

Resolution Must Be Sufficient

Representation resolution must be fine enough to preserve simulation-relevant differences.

For example:

If a safety event depends upon millisecond timing, a one-second temporal representation may be insufficient.

If a regional traffic simulation concerns aggregate traffic flow, centimeter-level road representation may be unnecessary. Resolution $\neq$ Depth

Simulation Depth asks how extensively the inquiry must examine the problem.

Representation Resolution asks how finely relevant reality distinctions must be represented.

The two must remain aligned but distinct.

Resolution Sufficiency

Possible representation-resolution outcomes may include:

Sufficient

Conditionally Sufficient

Partially Sufficient

Insufficient

or:

Undetermined Reality Simplification

Simulation frequently requires simplification.

A Reality Simplification removes or reduces complexity while attempting to preserve the characteristics necessary for the simulation inquiry.

Simplification may concern: geometry, interactions, human behavior, environmental variation, component structure, dependencies, system dynamics, or other dimensions.

Simplification Must Be Explicit

SIMULOS® establishes:

A material simplification must remain identifiable wherever it could affect

simulation meaning. Simplification ≠ Exclusion

A simplified element remains inside the simulation representation.

An excluded element remains outside the governed simulation inquiry.

Therefore:

Simplified ≠ Excluded

This distinction is important because an oversimplified representation and a formal exclusion create different governance consequences.

Reality Approximation

Some real-world characteristics cannot be represented directly and require approximation.

An approximation may use: surrogate relationships, estimated values, proxy behavior, statistical distributions, representative profiles, or other substitute mechanisms.

Approximation must remain explicit where material.

Proxy Representation

A representation may use a proxy for a real-world phenomenon.

For example:

Representative Human Behavior Model

may stand in for broader real human behavioral variability.

RRS requires the relationship:

Reality Target → Proxy Representation

to remain visible.

No Proxy Equivalence Rule

A proxy representation must not automatically be treated as equivalent to the reality it represents.

Representation of Dynamic Reality

Some reality changes continuously.

Examples include: adaptive AI systems, human populations, ecosystems, markets, autonomous systems, weather, organizational structures, and infrastructure networks.

RRS must establish whether the representation captures: static state, state

range, dynamic transition, adaptation, evolution, or another relevant form of change.

Static Representation of Dynamic Reality

A dynamic reality may sometimes be represented through a static approximation.

This may be legitimate for a narrowly bounded purpose.

However:

Static Representation ≠ Dynamic Equivalence

The resulting limitation must remain visible.

Representation of Human Behavior

Human behavior may be simplified into: behavioral profiles, probabilistic actions, bounded response models, cognitive approximations, scripted behavior, or agent-based representations.

RRS governs whether the selected human representation corresponds sufficiently to the simulation purpose.

It does not replace human cognition governance under CLIA®.

Representation of AI and Autonomous Behavior

AI and autonomous systems may require representation of: perception, decision behavior, planning, tool use, autonomy, interaction, constraints, memory effects, adaptation, or other capabilities.

RRS governs the simulation representation of those characteristics.

AIG®, ASGA™, CLIA®, and MGIA™ govern their respective underlying operational governance spaces.

Representation of Operational Environment

The Simulation Environment may itself represent selected operational reality.

Examples include: road conditions, industrial facilities, orbital environments, weather, infrastructure, networks, human populations, communication systems, or resource environments.

RRS establishes what parts of these conditions must be represented.

The later Simulation Model & Environment Standard governs how that representation is constructed and configured.

Representation Boundary

Every Reality Representation should have a defined boundary.

The Representation Boundary identifies:

which reality dimensions are represented

and:

which reality dimensions are not represented or are represented only indirectly. Object Boundary $\neq$ Representation Boundary

The Simulation Object may have a broader governed boundary than the portion represented inside one simulation.

Therefore:

Simulation Object Boundary $\neq$ Reality Representation Boundary

A mismatch is not automatically wrong.

It must be explicit and consistent with Simulation Scope.

Representation Coverage

Representation Coverage asks how much of the reality required by the governed simulation inquiry has actually been represented.

Possible statuses may include:

Complete for Governed Scope

Substantially Complete

Partial

Materially Incomplete

or:

Undetermined

Coverage refers to required reality within the simulation scope, not all reality.

Representation Gap

A Representation Gap exists where a reality element required by the governed Simulation Scope is not sufficiently represented.

Examples include: missing behavior, missing dependency, missing interaction, missing environmental condition, missing population variation, missing failure dynamics, or missing temporal behavior.

Representation gaps must remain visible.

Representation Distortion

A Representation Distortion exists where the simulation representation materially changes a real-world characteristic in a way capable of affecting the simulation inquiry.

Distortion may arise from: oversimplification, poor abstraction, inappropriate proxy behavior, excessive aggregation, inadequate temporal treatment, incorrect scaling, or unsuitable representation relationships.

Representation Bias

A representation may systematically favor certain states, behaviors, outcomes, or conditions.

Where such bias materially affects the simulation inquiry, it should remain governed as a Representation Limitation.

Representation Uncertainty

A simulation representation may contain uncertainty concerning how well the chosen representation corresponds to reality.

RRS does not require artificial certainty.

Representation uncertainty should remain explicit where material.

Representation Limitation

A Representation Limitation is a known condition under which the simulated representation does not fully reproduce a reality element relevant to the governed inquiry.

Limitations may result from: abstraction, simplification, unavailable data, incomplete knowledge, model capability, computational constraints, unresolved science, uncertain behavior, proxy use, or representational constraints.

Limitation Must Propagate

Material Representation Limitations must remain visible to downstream SIMULOS® governance.

Conceptually:

Representation Limitation → Assumptions → Model & Environment → Scenarios → Execution → Output Governance → Reality Correlation No Reality Equivalence Rule

SIMULOS® establishes:

A simulation representation must never be presumed equivalent to operational reality merely because the simulation executes successfully.

No Completeness-by-Detail Rule

High detail in represented elements does not establish that all material reality dimensions were represented.

No Fidelity-by-Complexity Rule

Greater representation complexity does not automatically establish greater fidelity.

No Resolution-by-Compute Rule

Greater computational scale does not automatically establish sufficient representation resolution.

No Hidden Simplification Rule

A material simplification must not remain invisible when it affects interpretation.

No Hidden Representation Gap Rule

A material reality element required by Simulation Scope must not disappear from the representation without becoming: a governed Representation Gap, an explicit exclusion, an assumption, or a returned scope question.

No Retrospective Representation Redefinition Rule

Reality Representation must not be materially recharacterized after simulation results are known merely to make results appear more realistic or broadly applicable.

Representation Change

Reality Representation may change as: new information becomes available, Simulation Scope changes, Applicability Context changes, Simulation Depth increases, object configuration changes, new failure modes emerge, or model capabilities improve.

Material Representation Change should remain traceable.

Representation Change Propagation

A material change to Reality Representation may affect: assumptions, model and environment design, input requirements, scenario design, execution, outputs, reality correlation, and lifecycle resimulation.

RRS identifies the changed representation basis and propagates its significance downstream.

Representation Versioning

Where Reality Representation materially changes, a distinguishable representation version may be required.

This preserves the relationship:

Simulation Object Version → Reality Representation Version → Simulation Activity Representation Status

Possible Reality Representation statuses may include:

Established

Conditionally Established

Partially Established

Representation Limited

Representation Conflicted

Representation Incomplete

or:

Representation Undetermined

Status communicates representation-governance condition.

It does not establish validation or operational reliance.

Reality Representation Record

RRS should support a persistent governed record capable of identifying: selected reality elements, abstraction levels, representation forms, fidelity requirements, achieved fidelity where assessed, resolution requirements, simplifications, approximations, proxies, representation boundaries, gaps, distortions, uncertainties, limitations, material changes, and representation status.

Operational Space

RRS governs: relevant reality selection, Reality Representation boundaries, abstraction, differentiated abstraction, Representation Fidelity, differentiated fidelity, Representation Resolution, simplification, approximation, proxy representation, dynamic reality representation, representation coverage, representation gaps, representation distortion, representation uncertainty, representation limitations, representation changes, representation versioning, and Reality Representation traceability.

Governance Boundary

RRS begins when:

a governed Simulation Object and Simulation Purpose & Scope basis exist and

the relevant reality must be translated into a form suitable for simulation.

RRS ends when:

the required reality elements have been sufficiently selected, abstracted, bounded, assigned representation fidelity and resolution requirements, simplified or approximated where necessary, and their material limitations documented to support governed Simulation Assumptions and Model & Environment construction.

Governance Scope

RRS governs only how selected reality is represented for simulation.

It does not govern: actual Operational Reality itself, Simulation Object identity, Simulation Purpose & Scope, general assumptions, model implementation, simulation-environment construction, input-data fitness, scenario design, simulation execution, output interpretation, formal validation, validation evidence sufficiency, general integrity, or operational reliance authorization.

These responsibilities belong to adjacent SIMULOS® spaces or complementary OOF® architectures.

Minimum Governance Requirements

A conforming application of RRS should establish:

1. the reality elements material to the governed Simulation Purpose and Scope, 2. an explicit Reality Representation boundary, 3. the selected representation form for material reality elements, 4. appropriate abstraction levels, 5. required Representation Fidelity, 6. required Representation Resolution, 7. material simplifications, 8. material approximations and proxies, 9. treatment of dynamic reality where relevant, 10. identification of Representation Gaps, 11. identification of material distortion or uncertainty, 12. explicit Representation Limitations, 13. prevention of reality-equivalence claims unsupported by the representation, 14. traceability of material Representation Changes, 15. and a persistent Reality Representation Record.

Governance Outputs

RRS may produce: Reality Selection Record, Material Reality Element Register, Reality Representation Basis, Reality Representation Boundary, Reality Representation Map, Representation Form Record, Abstraction Level Record, Differentiated Abstraction Map, Representation Fidelity Requirement, Representation Fidelity Profile, Representation Resolution Requirement, Spatial Representation Resolution, Temporal Representation Resolution, Behavioral Representation Resolution, Reality Simplification Record, Reality Approximation Record, Proxy Representation Record, Dynamic Reality Representation Record, Representation Coverage Assessment, Representation Gap Record, Representation Distortion Record, Representation Bias Record, Representation Uncertainty Record, Representation Limitation Record, Representation Change Record, Representation Version Record, Reality

Operational Applications

Reality Representation Standard may be applied across: artificial intelligence, autonomous systems, robotics, digital twins, aerospace, aviation, space missions, autonomous vehicles, industrial systems, manufacturing, infrastructure, healthcare simulation, scientific modeling, environmental modeling, financial simulation, population simulation, behavioral simulation, organizational simulation, mission planning, system-of-systems simulation, training environments, critical infrastructure, and future operational ecosystems.

Relationship to Other OOF® Architectures

Reality Representation Standard interoperates with: GOA™ ORA™ VALIDOS® INTEGROS® ADIT® AGA™ AIG® CLIA® MGIA™ ASGA™ RIS™

while preserving a strict architectural boundary.

ORA™ governs Operational Reality itself.

RRS governs:

how selected Operational Reality is represented within simulation.

This distinction is fundamental:

ORA™: What is actually occurring?

RRS: How is the relevant reality represented for simulation?

VALIDOS® may subsequently determine whether a simulation, simulation-derived claim, or simulation evidence is sufficiently validated for a defined purpose.

RRS does not make that Validation Determination.

INTEGROS® may govern the integrity of representation records, data, models, and related objects.

RRS governs their simulation-specific representational role.

ADIT® may preserve evidence demonstrating how a representation was constructed, changed, or executed.

AIG®, CLIA®, MGIA™, and ASGA™ may define domain-specific characteristics that a simulation representation must reflect when AI, cognition, memory, or autonomous behavior is being simulated.

RRS consumes those characteristics without replacing their governance architectures.

Architectural Position

Reality Representation Standard (RRS) is the third Parent Standard of SIMULOS®
— Simulation Governance Architecture.

The architectural progression is:

Simulation Object

↓

Simulation Purpose & Scope

↓

Reality Representation

SOS establishes:

What exactly is being simulated?

SPSS establishes:

Why is it being simulated, what must it cover, and within which contextual and depth boundaries?

RRS establishes:

Which parts of reality must be represented and how must that representation remain bounded by abstraction, fidelity, resolution, simplification, approximation, and known limitations?

Once the Reality Representation Basis has been established, SIMULOS® can proceed to the next architectural question:

Which assumptions are required to make that representation and the resulting simulation possible, and what limitations do those assumptions introduce?

This transition leads directly into:

Simulation Assumption Governance

Foundational Principle

A simulation can legitimately represent reality only to the extent that the reality elements material to its purpose are explicitly selected, appropriately abstracted, sufficiently represented, and bounded by their known representational limitations.

Canonical Closing Statement

Reality Representation Standard (RRS) establishes the representational bridge between Operational Reality and governed simulation by defining which aspects of reality must be selected, abstracted, simplified, approximated, resolved, and represented for the intended Simulation Purpose and Scope. Through governance of Reality Selection, abstraction, fidelity, resolution, dynamic

representation, proxies, coverage, gaps, distortion, uncertainty, limitations, change, and traceability, RRS prevents simulation representations from being mistaken for reality merely because they are technically detailed or successfully executed. By making the transformation from reality to representation explicit and governable, RRS establishes one of the central methodological foundations of SIMULOS® and provides the reality basis upon which assumptions, models, environments, inputs, scenarios, execution, outputs, and later simulation-to-reality correlation can legitimately be constructed.

Module Architecture

→ [Reality Selection Module \(RSM\)](#)

→ [Reality Abstraction Module \(RAM\)](#)

→ [Representation Fidelity Module \(RFM\)](#)

→ [Representation Resolution Module \(RRM\)](#)

→ [Representation Limitation Module \(RLM\)](#)

Related Documents

→ [About Reality Representation Standard \(RRS\)](#)

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Canonical Source

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